

56)



$$B = \frac{\mu_0 i}{2R} \rightarrow \text{campo no eixo de um rallo} \quad B = \frac{\mu_0 N i}{2} \frac{R^2}{(R^2 + x^2)^{3/2}} \rightarrow \text{eixo N rallo} \rightarrow B = \frac{\mu_0 N i}{2R}$$

$$a) \frac{B_a}{B_b} = \frac{\frac{\mu_0 i}{2R}}{\frac{\mu_0 i}{2R}} = \frac{R}{R} = 1$$

$$b) \mu = N i A \rightarrow \frac{\mu_a}{\mu_b} = \frac{N_a i_a A_a}{N_b i_b A_b} = 1$$

Prova orlgo:

2) $q = +2 \cdot 10^{-3} \text{ C}$ $F_{\text{mag}} = 20 \cdot 10^{-6} \text{ N}$ a) O campo está entrando no plano onde está o laço e a corrente tem que estar para o sentido.

$v = 100 \text{ m/s}$ $\mu_0 = 4 \cdot \pi \cdot 10^{-7} \text{ T} \cdot \text{m/A}$ b) $\frac{4 \cdot \pi \cdot 10^{-7} \cdot i}{2 \cdot \pi \cdot 0.1} = 10^{-4} \rightarrow i = \frac{2 \cdot \pi \cdot 0.1 \cdot 10^{-4}}{4 \cdot \pi \cdot 10^{-7}} = 50 \text{ A}$

$20 \cdot 10^{-6} = 2 \cdot 10^{-3} \cdot 100 \cdot B \cdot \sin 90^\circ \rightarrow B = 10^{-4} \text{ T}$

3) Solenóide: $i = 2 \text{ A}$ $N = 100$ $l = 3.14 \text{ m}$ $\pi = 3.14$ $B_s = \frac{4 \cdot \pi \cdot 10^{-7} \cdot 2 \cdot 100}{3.14} = 8 \cdot 10^{-5} \text{ T}$

campo resultante do laço $\rightarrow B_{\text{res}} = 8 \cdot 10^{-5} \text{ T}$

a) $\vec{B}_R = \vec{B}_{\text{ext}} + \vec{B}_m \rightarrow 8 \cdot 10^{-5} = 8 \cdot 10^{-5} + \vec{B}_m \rightarrow \vec{B}_m = 4 \cdot 10^{-6}$ $B_m = \chi_m \cdot B_{\text{ext}} \rightarrow \chi_m = \frac{4 \cdot 10^{-6}}{8 \cdot 10^{-5}} = 0.05$ Suscetibilidade magnética

b) $M = \frac{\chi_m}{\mu_0} \cdot B_{\text{ext}} \rightarrow M = \frac{0.05}{4 \cdot \pi \cdot 10^{-7}} \cdot 8 \cdot 10^{-5} = 3.18 \text{ A/m}$

Lista 1:

5) $B = B_x \hat{i} + 3B_y \hat{j}$ $v = 2\hat{i} + 4\hat{j}$

$$F = 6.4 \cdot 10^{-19} \hat{b}$$

$$F = q v \times B = 6B_x \hat{b} - 4B_y \hat{a} = 2B_x \hat{b}$$

$$6.4 \cdot 10^{-19} = 2 \cdot 1.6 \cdot 10^{-19} \cdot 2 \times B$$

$$B_x = 2 \text{ T}$$

1) $e = -1.6 \cdot 10^{-19}$ $v_x B = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0.03 & -0.15 & 0 \\ 2 \cdot 10^6 & 3 \cdot 10^6 & 0 \end{vmatrix} = 0.03 \cdot 3 \cdot 10^6 - 0.15 \cdot 2 \cdot 10^6 = 0.09 \cdot 10^6 + 0.3 \cdot 10^6 = 3.9 \cdot 10^5$

$$F = -1.6 \cdot 10^{-19}$$

8) $V = \frac{E}{B} \rightarrow v = \frac{1.5 \cdot 10^3}{0.4} = 3750$ p 1 metro de comprimento

18) $v_0 = 0$ $B = 200 \cdot 10^{-3} \text{ T}$ $\Delta U = 350 \text{ V}$ b) $R = \frac{9.1 \cdot 10^{-31} \cdot 11 \cdot 10^6}{1.6 \cdot 10^{-19} \cdot 200 \cdot 10^{-3}} = 3.12 \cdot 10^{-4} \text{ m}$

$$R = \frac{m \cdot v}{q \cdot B}$$

$$E_{\text{el}} = q \Delta U$$

$$E_{\text{cin}} = \frac{m v^2}{2}$$

a) $q \Delta U = \frac{m v^2}{2} \rightarrow v = \sqrt{\frac{2 q \Delta U}{m}} = 11 \cdot 10^6 \text{ m/s}$

24) $F = q v \times B \rightarrow F = q \cdot v \cdot B \cdot \sin 90^\circ \rightarrow 3.2 \cdot 10^{-20} = 1.6 \cdot 10^{-19} \cdot v \cdot 4 \cdot 10^{-3} \rightarrow v = 5 \cdot 10^6 \text{ m/s}$

b) $R = \frac{m \cdot v}{q \cdot B} = \frac{9.1 \cdot 10^{-31} \cdot 5 \cdot 10^6}{1.6 \cdot 10^{-19} \cdot 4 \cdot 10^{-3}} = 7 \cdot 10^{-3} \text{ m}$ $T = \frac{2 \cdot \pi \cdot R}{v} = 8.8 \cdot 10^{-9}$

27) $F_{\text{mag}} = q v B \rightarrow v_y = \frac{F_{\text{mag}}}{q B} = 4.17 \cdot 10^4 \text{ m/s}$

$p = \frac{2 \cdot \pi \cdot m \cdot v_x}{q \cdot B} \rightarrow v_x = 5.03 \cdot 10^4 \text{ m/s}$ $v_k = \sqrt{v_x^2 + v_y^2} = 6.53 \cdot 10^4 \text{ m/s}$

* Tem velocidade em x e y, portanto, tem como fazer para obter o resultado.

39) $P = F_m \rightarrow m \cdot g = i \cdot L \cdot B \cdot \sin 90^\circ \rightarrow 73 \cdot 10^{-2} = i \cdot 0.62 \cdot 0.44 \rightarrow i = 0.47 \text{ A}$, para o dentro

41) $i = 5000$ $B = 60 \cdot 10^{-6}$ $\theta = 70$ $L = 100 \text{ m}$ $F = i \cdot L \cdot B \cdot \sin \theta \rightarrow F = 5000 \cdot 100 \cdot 60 \cdot 10^{-6} \cdot \sin 70 = 28.2 \text{ N}$

(44) $F = i \cdot L \cdot B \cdot \sin \theta \rightarrow F = 9,13 \cdot 10^{-3} \cdot 2,56 \cdot 10^{-2} \cdot 56,3 \cdot 10^{-3} \approx 1,31 \cdot 10^{-5}$

$F = m \cdot a \rightarrow a = \frac{F}{m} = \frac{1,31 \cdot 10^{-5}}{2,56 \cdot 10^{-2}} = 5,08 \cdot 10^{-4}$

$v = v_0 + a \cdot t = 5,08 \cdot 10^{-4} \cdot 6,71 \cdot 10^{-3} = 3,41 \cdot 10^{-6} \text{ m/s}$

(49) $R = 1,8 \cdot 10^{-2} \text{ m}$

$F = i \cdot L \cdot B \cdot \sin \theta$

(54) $\psi = N \cdot i \cdot A \rightarrow \psi = 1,2 \cdot 6 \cdot \pi \cdot 10^{-2} = 0,18 \text{ T}$

(56) $8,10^{22} = 1 \cdot i \cdot \pi (3500 \cdot 10^{-3})^2$

$B = 3,4 \cdot 10^{-3} \text{ T}$

$F = 4,6 \cdot 10^{-3} \cdot 2,56 \cdot 10^{-2} \cdot 3,4 \cdot 10^{-3} \cdot \sin 20^\circ$

$\tau = \psi \cdot B \cdot \sin \theta \rightarrow \tau = 0,181 \cdot 12 \cdot \sin 41^\circ$

$i = 20 + 8758440 \text{ A}$

Lista 2:

(32) $E_{\text{final}} = 4000 \text{ eV}$

* Último ganho energia

$\Delta x = v \cdot \lambda$

$E_n = e \cdot 2000 = 2000 \text{ eV}$

$\lambda = \frac{h \cdot c}{E} = \frac{1240}{2000} = 0,62 \text{ nm}$

Primeira região:

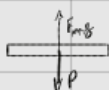
Segunda região:

$T = \frac{2 \cdot \pi \cdot m}{q \cdot B} = \frac{2 \cdot \pi \cdot m}{q \cdot 0,01} = 3,57 \cdot 10^{-9} \text{ s}$

$E = 3,6 \cdot 10^{19} \text{ eV}$

$E_T = \frac{m \cdot v^2}{2} \rightarrow v = \sqrt{\frac{2 \cdot E_T}{m}} = 45,9 \cdot 10^6 \text{ m/s}$

(39)



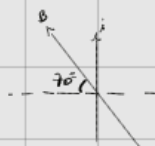
$i \cdot B \cdot L \cdot \sin 90^\circ = m \cdot g$
 $\rightarrow i = \frac{m \cdot g}{B \cdot L \cdot \sin 90^\circ} = \frac{13 \cdot 10^{-3} \cdot 10}{0,99 \cdot 6 \cdot 10^{-2}} = 2,1 \cdot 10^{-2} \text{ A}$

(44)

$F = i \cdot L \cdot B \cdot \sin \theta \rightarrow F = 9,13 \cdot 10^{-3} \cdot 2,56 \cdot 10^{-2} \cdot 56,3 \cdot 10^{-3} \cdot \sin 90^\circ$

$F = 1,31 \cdot 10^{-5} \text{ N}$
 $F = m \cdot a \rightarrow a = \frac{F}{m} = \frac{1,31 \cdot 10^{-5}}{2,56 \cdot 10^{-2}} = 5,08 \cdot 10^{-4}$

(47)



$F = i \cdot L \cdot B \cdot \sin \theta \rightarrow F = 5000 \cdot 100 \cdot 60 \cdot 10^{-6} \cdot \sin 70^\circ$

$F = 28,2 \text{ N}$

b) Para cima, mas no plano perpendicular

$v = v_0 + a \cdot t \rightarrow v = 5,08 \cdot 10^{-4} \cdot 6,71 \cdot 10^{-3} = 3,41 \cdot 10^{-6} \text{ m/s}$

(44)

$\tau = N \cdot i \cdot A \cdot B \cdot \sin \theta$

$\tau = 20 \cdot 0,1 \cdot 0,5 \cdot 10 \cdot 30 = 0,5$

COM O NORMAL

(11)

$i = 10 \text{ A}$

$d = 10 \cdot 10^{-3} \text{ m}$

distância de b

$B = \frac{\mu_0 \cdot i}{2 \cdot \pi \cdot r} \rightarrow \text{campo de cada fio}$

$B_1 = B_2 = B = \frac{\mu_0 \cdot i}{2 \cdot \pi \cdot r} = \frac{4 \cdot \pi \cdot 10^{-7} \cdot 10}{2 \cdot \pi \cdot 10^{-2}} = 2 \cdot 10^{-5} \text{ T}$

(24)

$B = \frac{\mu_0 \cdot i}{2 \cdot \pi \cdot r} \quad R = 9 \cdot 10^{-2}$

Situação a: Se somo

$\frac{\mu_0 \cdot i}{4 \cdot \pi \cdot R} + \frac{\mu_0 \cdot i}{4 \cdot \pi \cdot r} = 47,25 \cdot 10^{-6}$

$\frac{\mu_0 \cdot i}{4} \left(\frac{1}{R} + \frac{1}{r} \right) = 47,25 \cdot 10^{-6}$

Situação b: Se subtrai

$\frac{\mu_0 \cdot i}{4 \cdot \pi \cdot R} - \frac{\mu_0 \cdot i}{4 \cdot \pi \cdot r} = 15,75 \cdot 10^{-6}$

$\frac{\mu_0 \cdot i}{4} \left(\frac{1}{R} - \frac{1}{r} \right) = 15,75 \cdot 10^{-6}$

$\begin{cases} \frac{K}{R} - \frac{b}{r} = 15,75 \cdot 10^{-6} \\ \frac{b}{R} + \frac{K}{r} = 47,25 \cdot 10^{-6} \end{cases}$

$\frac{b}{R} = 6,3 \cdot 10^{-6}$